Article

Psychedelics broaden the repertoire of brain states

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Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience.

Introduction

Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time.

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Results

Entropy increases under psilocybin

We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time.

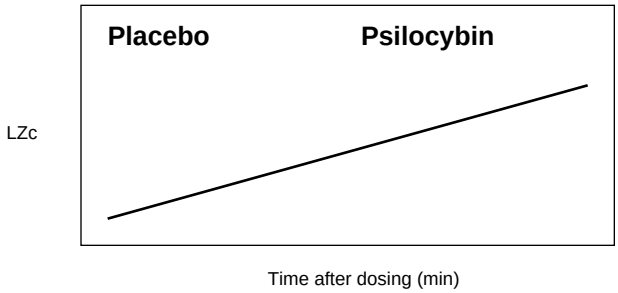

Fig. 1 | Signal complexity across conditions. Lempel–Ziv complexity rose after dosing.

Subjective reports track entropy

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Discussion

Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups.

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References

1. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2001 (2019).
2. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2002 (2019).
3. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2003 (2019).
4. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2004 (2019).
5. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2005 (2019).
6. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2006 (2019).
7. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2007 (2019).
8. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2008 (2019).
9. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2009 (2019).
10. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2010 (2019).
11. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2011 (2019).
12. Author, A. et al. A study of entropy in the brain. J. Neurosci. 2012 (2019).

Methods

Participants

Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise.

Statistical analysis

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Data availability

Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions.

Acknowledgements

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Author contributions

Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise.

Competing interests

The authors declare no competing interests.